

Fig. 5: Actual-size PCB layout for the Arduino FM receiver

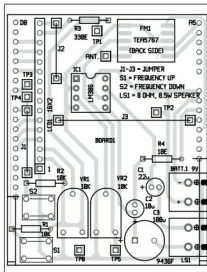


Fig. 6: Component layout for the PCB

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com

operation is for search-up or search-down operation. The implementation of these operations has been included in the code.

R1 and R2 are pull-up resistors connected at digital pins 7 and 8 of Arduino board, respectively.

Construction and testing

An actual-size (Arduino shield type), single-side PCB for Arduino FM receiver is shown in Fig. 5 and its component layout in Fig. 6. Assemble the components on the recommended PCB to avoid assembly errors. Double check for any overlooked error. TEA5767 is to be mounted on solder-side of the PCB.

A 75cm (wavelength/4) hookup wire antenna should be connected at pin 10 of the TEA5767 module.

Switch on the circuit and use switch S1 or S2 to change frequency. Pressing S1 momentarily increases the frequency and S2 decreases the frequency. Long pressing of any of the two switches will enable auto scanning.

LCD1 shows the selected frequency, signal strength and whether the sound is mono or stereo.

To test the circuit for proper functioning, first verify the correct voltages at various points as given in the test points table. ●

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```
Bitwise operators
& (bitwise and)
| (bitwise or)
^ (bitwise xor)
~ (bitwise not)
<< (bitshift left)
>> (bitshift right)
```

Since structures of the first and the second bytes are different, different bitwise operators are used.

Read mode. In read mode, we have to note the following changes:

1. When a station is found in search mode, as instructed during write operation, the ready flag (RF) in first byte is set to 1. That is, RF=1
2. The values of signal strength level in 4th byte
3. Stereo or mono reception in 3rd byte

For reading the values from TEA5767, the following method is used in the code by creating a buffer as follows:

```
Buffer[i]= Wire.read ();
Wire.read (); is used to read the values of all the five bytes in Read mode.
```

Displaying values on LCD screen. The task of displaying values is performed by reading the values of bytes of the TEA5767 in read mode.

1. The available frequency is displayed on LCD screen by reading first byte stored in buffer(0); this value is stored in buffer during read operation
2. If search mode is high, a scan is displayed on the screen
3. The signal strength is stored in the 4th byte which can be obtained by reading the buffer(3)
4. The reception type is displayed on screen by reading the buffer(3).

Searching/changing the frequency. Switches S1 and S2 are used to change the station/channel search or switch to auto-scan mode. Search-up or search-down operation depends on the switch pressed. The 7th bit of 3rd byte in write

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